



Effects of Failure Types on Trust Repairs in Human–Robot Interactions

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Abstract

System performance is the central determinant of user trust in human–machine communication; however, performance failure is inevitable. This study develops a three-fold typology of performance failures (i.e., logic, semantic, and syntax) commonly observed in human-robot interactions based on the differences between the expected and actual outcomes. Herein, 1027 observations are collected from an online experiment to elucidate how the three types of failure and four repair methods (namely, internal attribution apology, external attribution apology, denial, and no repair) impact user trust while examining blame attributions as an underlying mechanism. The results reveal that despite some similarities, the interactions between trust violation types and repair methods differ in robot-to-human trust repair from those in human-to-human trust repair, which contradicts previous findings. Logic failures are found to be the most detrimental category of performance failures, and the internal-attribution apology is the optimal repair strategy. Notably, participants report greater levels of competence-based trust beliefs if they believe that the situation is jointly controlled by the human interactant and robot.

Keywords Performance failures · Trust repairs · Blame attributions · Human–robot interactions

1 Introduction

Robots have gradually become an indispensable part of modern life and are involved in numerous essential aspects of production, distribution, transportation, retail, and services. More recently, the global COVID-19 pandemic has significantly increased robot popularity owing to concerns regarding the risk of human-to-human contact, and this trend is expected to continue post-pandemic [1, 2]. With increasing human interactions with robotics, both proximal and remote

[3], developing and maintaining human-to-robot trust has become crucial as trust is a critical determinant of technology adoption and use [4]. Scholars across multiple disciplines have long been interested in exploring factors that boost or hinder human-to-robot trust in human-robot interactions (HRIs).

Robotic performance (i.e., the process or action of robots executing assigned tasks) is the central determinant for human-to-robot trust [5–7]. However, many limitations need to be overcome to realize optimal performance in robots. Moreover, the designated actions of robots are inevitably disrupted by all types of failures, including inaccurately decoding voice commands, failing to recognize presented objects, and incorrectly planning routes. Such robotic failures and errors can generate significant negative impacts on user trust, as delineated by previous studies [8–10]. Robotic failures are unpleasant but inevitable. In addition to the continuous pursuit of technological advancement, the critical question is how to restore or rebuild human-to-robot trust under circumstances in which robots make mistakes. After proposing an output-oriented HRI failure typology (i.e., logic, semantic, and syntax failures) and reviewing previous research on human-to-human trust repair [11, 12], the present study specifically focuses on trust repair in HRIs and explores how different failure types and repair strategies (i.e., internal

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attribution apology, external attribution apology, denial, and no repair) may result in different repair outcomes. Additionally, this study investigates blame attributions as a potential underlying mechanism in HRI trust repairs.

2 Trust Violations and Repairs

Trust is a desideratum for effective and efficient collaboration in every situation involving risks [13], regardless of whether it is a human-to-human interaction or HRI. Although humans and robots are ontologically different, the concept of trust is applicable to humans relationships with both other humans and robots [14]. This concept has been extensively investigated across multiple disciplines; by 2013, over 300 definitions of trust were documented, as summarized by Schaefer [15]. Trust, traditionally defined as a psychological construct emerging from human-to-human interactions, has been reconsidered based on the prevalence of HRI [16]. People's willingness to rely on an agent is based on whether they believe that the agent possesses desirable attributes [17]. Following Lee and See [18], this study defines trust as “the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability” (p. 54).

Fundamentally, trust is closely linked to the expectation that trustors develop on trustees. Mayer et al. [19] identified three perceived characteristics of trustees: ability (i.e., “skills, competencies, and characteristics that enable the party to have influences within some specific domain”, p. 717), benevolence (i.e., “the extent to which a trustee is believed to want to do good for the trustor”, p. 718), and integrity (i.e., “the trustor's perception that the trustee adheres to a set of principles that the trustor finds acceptable”, p. 719). People develop expectations of the three aforementioned characteristics and thus form three types of trust: competence-, benevolence-, and integrity-based trust. Based on a meta-analysis conducted by Malle and Ullman [20], human trust in robots also shares similar dimensionality, akin to human-human trust. They argue that the conceptualizations of HRI-based trust can be categorized into two basic aspects, performance (i.e., competence and reliability) and morality (i.e., sincerity, integrity, and benevolence), and showed empirical evidence supporting this bifactor structure. Following their proposition, the current study also assumed a bifactor structure of human-robot trust. Herein, competence- and integrity-based trust, the two most frequently examined trust dimensions in both human-human and human-robot trust research, is investigated.

When such expectations are not met, trust violations happen as a result of “unmet expectations concerning another's behavior, or when the person does not act consistent with one's values” (p. 248) [21], and trust violations can lead

to trust loss. To avert the social and economic loss accompanying the loss of trust [22], trustees may take action to restore positive expectations while reducing negative ones [23], namely, trust repair attempts. Considering the importance of trust development and ubiquity of trust violations in practice, research on factors that can potentially influence trust repair outcomes is ongoing. Human-to-human trust research is an important reference point for human-to-robot trust research. We briefly reviewed trust repair research in human-to-human communication to fully understand the discussion of trust repairs in HRIs.

2.1 Human-to-Human Trust Repairs

Research on trust repair began in the early 2000s and comprised a series of studies conducted in organizational settings. In a classic study by Kim et al. [11], job interview scenarios were set, where participants were instructed to watch videotapes of job candidates who demonstrated either competence- or integrity-based trust and exercised either apology (i.e., a response in which the trustee accepts responsibility, expresses repentance, and stresses the intent to avoid similar violations in the future) or denial (i.e., a response in which the trustee avoids responsibility and expresses no repentance) to repair trust. Their study aimed to investigate how the interaction between violation types and repair strategies may influence the trajectory of trust restoration and development.

They recognized perceived competence and integrity as two determinants of trust [19], and referred to the concept of hierarchically restrictive schemas in cognitive psychology for their theoretical explanations [24]. Hierarchically restrictive schemas propose that people's evaluations of intelligence and morality are asymmetrical in that an incident signaling inability tends to be considered an anomaly, whereas a single dishonest behavior is considered typical [24]. This is because people believe that individuals with negative morality traits or positive intelligence traits have a wider range of behaviors at their disposal than those with positive morality traits or negative intelligence traits [25]. An intelligent person may accidentally fail, but a moral person never does anything unethical. Consequently, integrity-based trust violations are generally considered to be harsher than competence-based ones. Thus, trust repairs should highlight positive information for competence-based trust violations because positive information outweighs negative information in intelligence-based schemas, while the key to repairing trust after integrity-based violations is to minimize negative information, given that negative information outshines positive information [11].

Based on the deductions above, Kim et al. [11] surmised that an apology would outperform denial for competence-based trust violations because positive information (i.e.,

expressions of repentance and intentions to avoid similar violations in the future) in an apology is emphasized over negative information (i.e., acceptance of culpability) when trust violations are competence-based. However, when violations are integrity-based, denial should be a better solution, as it precludes negative information by denying responsibility. Their data primarily supported the proposed interaction between trust violation type and repair strategy.

Subsequent studies have extended this framework. Kim et al. [12] further divided apology into internal and external attribution apology based on attributions of causality and found that internal attribution apology (i.e., apology assuming full responsibilities and attributing blame to internal factors) restores more trust than external attribution apology (i.e., apology assuming partial responsibilities and attributing the rest to external factors) for competence-based violations, whereas vice versa is true for integrity-based apology. Ferrin et al. [26] introduced reticence (i.e., a response in which the trustee claims that they cannot or will not confirm or deny responsibility) into the comparisons, noting that the effectiveness of reticence equates denial under competence-based trust violations and apology under integrity-based violations. Kim et al. [27] promoted this framework from dyadic interactions to small-group interactions, essentially maintaining interaction effects in addition to the strong influence of social context. Recently, this framework has been repeatedly verified in various organizational settings [28–33].

However, not all studies have reported similar interactions. For example, Utz et al. [34] found that an apology is more effective than denial for both competence- and integrity-based violations in the context of online transactions. Similarly, Bansal et al. [35] observed better trust repair outcomes following an apology rather than denial for competence-, benevolence-, and integrity-based trust violations in the context of privacy breaches. Such exceptions present potential challenges to the framework proposed by Kim et al. [11] and require further investigation.

2.2 Robot-to-Human Trust Repairs

As performance failures are inevitable in HRIs, scholars have been attempting to develop solutions for trust violations caused by nonoptimal performance. Various trust repair strategies have been tested in diverse contexts, including human-automation, human-agent, and human-robot interactions, such as blaming [36, 37], apologizing [38–45], denying [39, 41, 43], promising [42, 45, 46], justifying [47], gaslighting [39], engaging in social dialogues [48], offering palpable compensations [40], offering options [40], and providing additional information [42] or support [49]. In general, the literature indicates that trust repair attempts are effective, regardless of the strategy used, but the investigation has been fragmented, lacking systematical approaches.

To date, two studies have tested Kim et al. [11] framework of trust repair in the context of human–machine interaction. Quinn [41] investigated the interaction between failure types (competency and integrity failures) and repair strategies in the context of self-driving automation systems, the results of which partially support this framework. Trust in apologetic automation was found to be greater following competence-based failures compared with integrity-based failures. However, trust in denying automation did not significantly differ between the two failure types, which challenges the existing framework.

Sebo et al. [43] examined the interactions between two trust violation types (competence- or integrity-based) and two trust repair methods (apology and denial). In the context of a competitive game, the experimental robot was programmed to break the promise set at the beginning. Half-way through the game, the robot framed the trust violation as either competence or integrity deficiency and repaired trust with either an apology or denial, which resulted in four message variants: (a) competence apology, (b) competence denial, (c) integrity apology, and (d) integrity denial. Measuring trust with participants' gaming choices during the latter half of the game and the Robotic Social Attributes Scale, Sebo et al. [43] found that when the violation was framed as a competence issue, apology outperformed denial; however, when it was framed as a matter of integrity, denial outperformed apology. This finding is similar to the interaction between trust violation types and repair methods found in human-to-human interactions, thus suggesting that the latter could be applied to HRIs. Nevertheless, the effects of failure types and trust repair methods still need to be further explored in HRIs owing to the scarcity of relevant research and inconsistent findings.

To gain a better understanding of trust repair in HRIs, the present study introduced two variants of apology, in addition to denial: internal and external attribution. These two variants differ in the framing of the locus. Essentially, internal attribution apology (e.g., “Sorry, I was too timid to ask questions”) fully accepts responsibilities by attributing causes of failures to internal factors, whereas external attribution apology (e.g., “Sorry, the question was phrased too ambiguously”) assumes only partial responsibilities by blaming external factors. In past human-to-human trust research, external attribution apology was found to be less effective than internal attribution apology but more effective than denial against competence-based trust violations; whereas it was less effective than denial but more effective than internal attribution apology against integrity-based trust violations [12]. However, the division between internal and external attribution apologies has not yet been investigated in the context of HRIs.

2.3 Categorizations of Performance Failures

“Failures” and “errors” were often used interchangeably in the past literature. To differentiate the two, the current study defines “failures” as “a degraded state of ability which causes the behavior or service being performed by the system to deviate from the ideal, normal, or correct functionality” (p. 9) [49]. Essentially, this definition emphasizes the negative impacts of errors (i.e., system flaws) on user experience if identified as negative violations of expectations. Several scholars have developed typologies to categorize various performance failures. For example, based on the locus of faults, Parasuraman et al. [50] categorized failures by locating faults, including information acquisition, information analysis, decision and action selection, and action implementation. Carlson and Murphy [51] classified failures as interactions, algorithms/methods, software design/implementation, and hardware failures. Based on the type of violated expectations, Giuliani et al. [52] divided failures into technical failures and social-norm violations. Based on the mechanisms of errors, numerous studies have categorized errors as omission and commission errors (e.g., Davenport and Bustamante [53]; Johnson et al. [54]; Madhavan et al. [55]; Rovira and Parasuraman [56]). Based on the occurrence patterns of failures, Ferrell [57] categorized robotic failures into individual, concurrent, and accumulative failures. According to the severity of the aftereffect, Carlson and Murphy [51] categorized failures as noncritical, repairable/compensable, and terminal failures. Based on recoverability of failures, Ross et al. [58] identified anticipated, exceptional, and unrecoverable errors.

Nonetheless, two potential issues exist in current failure typologies. First, several typologies categorize failures according to their technical causes resulting from software and hardware problems. However, lay users are not trained to precisely identify such causes [59]; rather, they judge performance failures primarily based on what is immediately available to them, i.e., the output generated by the machine. Therefore, failure categorizations must be more human-centered to facilitate HRI research. Second, multiple typologies classify failures by overall failure outcomes, yet in some situations, the outcomes generated by different failures do not differ considerably. Under such circumstances, extant typologies cannot fully capture the nuances of the diverse failure types and their consequences.

Drawing on error typologies in computer science [60], this study proposes an output-oriented failure typology. Focusing on the relationships between the expected output and actual output, the current study categorizes performance failures into three types: logic, semantic, and syntax failures. Logic failures produce something relevant but wrong; for example, answering “ $1 + 1 = ?$ ” with “3”. Semantic failures produce something completely irrelevant; for example, answering “ $1 + 1 = ?$ ” with “Apple”. Syntax failures produce nothing; for

example, answering “ $1 + 1 = ?$ ” with silence. Altogether, this three-fold typology covers all performance failures in HRIs across various contexts.

Owing to the lack of relevant research and theoretical guidance, predicting how people perceive the three types of failure in a robot, determining the optimal trust repair strategy, and determining whether an interaction between failure types and repair strategies exist are challenging. Therefore, this study asked the following research questions.

RQ1 How do different types of performance failure affect competence- and integrity-based trust beliefs, and the perceived severity of violations in HRIs?

RQ2 How do the different types of trust repair attempts impact competence- and integrity-based trust beliefs, and the perceived severity of violations in HRIs?

Research on both human-to-human and human-to-machine trust has clarified that different trust repair methods exert distinct effects on the various types of trust violations [11, 41, 43]. Therefore, this study asked whether any specific combination of failure types and repair strategies may generate significant effects on people’s trust in robots.

RQ3 How do failure types and trust repair methods interact to influence competence- and integrity-based trust beliefs, as well as perceived severity of violations in HRIs?

3 Attribution as Underlying Mechanisms

3.1 Causal Attributions in Trust Repairs

Attribution theory is one of the most salient theoretical perspectives in trust repair research [61, 62]; it has been applied in multiple studies to explain the effects of trust repairs (e.g., Bansal et al. [35]; Goles et al. [63]; Quinn [41]). According to Weiner [64], people assign explanations for behaviors and events in three dimensions: locus (i.e., whether perceived causes are located in external situational factors or the actor’s internal qualities), stability (i.e., whether perceived causes are fluctuant or constant), and controllability (i.e., whether the perceived control of reinforcement is external or internal). These perceptions can stimulate more positive interpretations that failures are caused internally by trustors themselves rather than by external factors (i.e., situations or trustees), failures are occasional, and failures cannot be controlled (e.g., Korsgaard et al. [65]; Shaw et al. [66]; Stouten et al. [67]; Takaku [68]). Therefore, better trust repair outcomes should be achieved if participants believe that: (a) their internal qualities (i.e., internal locus) rather than the robot’s

innate qualities or situational factors are the causes of failures; (b) the causes are unstable over time (i.e., instability); and (c) the mistake is uncontrollable by the robot (i.e., external uncontrollability) but controlled by the human interactant (i.e., personal controllability).

H1 After failure occurs, competence-based trust in robots is better repaired when more (a) internally, (b) unstable, and (c) externally uncontrollable and/or personally controllable causal attributions are made in HRIs.

H2 After failure occurs, integrity-based trust toward robots is better repaired when more (a) internally, (b) unstable, and (c) externally uncontrollable and/or personally controllable causal attributions are made in HRIs.

H3 After failure occurs, the perceived severity of violations by robots is lower when more (a) internally, (b) unstable, and (c) externally uncontrollable and/or personally controllable causal attributions are made in HRIs.

3.2 Trust Repair as Blame Manipulations

Different trust repair strategies can be approached as different manipulations of causal attributions. The latent logic in trust repair is that one end of each attribution continuum tilts the other [69]. For instance, if one makes more unstable attributions in this case, they will naturally reduce stable attributions, thereby decreasing the attribution of guilt. This alleviates the negative effects of violations on perceived trustworthiness by ruling out the notion that these failures are caused by deficiencies in ability or integrity [62].

Under the scope of blame attributions, denial (e.g., “It was not my fault”) asserts external and uncontrollable attributions [5, 41], whereas apology typically weakens stability attributions by portraying changeability (e.g., “I promise to do better in the future”), although provoking internal and controllable attributions by expressing repentance (e.g., “I am very sorry”). Compared with internal attribution apology (e.g., “Sorry, I did not pay attention”), which makes complete internal and controllable attributions, external attribution apology partially rejects internal attributions by redirecting blame to external and uncontrollable factors (e.g., “Sorry, the room is too noisy”).

Following the logic of blame attributions, denial should result in more external and uncontrollable attributions than an external attribution apology, with an internal attribution apology being the least favored because of greater internal and controllable attributions. However, a contradiction arises because an apology should outperform denial owing to greater levels of unstable attributions entailed by promises for the unknown future. Moreover, external attributions in trust repair also entail potential risks because making excuses can be perceived as deceptive, self-absorbed, and ineffectual

[70], thus diminishing trustors’ willingness to reconcile [71]. As people dislike lying robots [72], external attribution and denial can be counterproductive when trustors are not convinced of robots’ innocence. Therefore, the results of trust repairs are influenced by the actual attributions trustors make. Hence, the current study explored the effects of failure types and repair methods on blame attributions by addressing the following questions.

RQ4 How do different types of performance failure impact blame attributions in HRIs?

RQ5 How do different types of trust repair attempts affect blame attributions in HRIs?

RQ6 How do failure types and trust repair methods interact and influence blame attributions to HRIs?

3.3 Entity Beliefs as a Contributor

In addition to the classic attribution theory, another important framework illustrates the effects of people’s beliefs on human attributes on their judgments and reactions in blame attribution [73–75]: implicit theories of moral responsibility. These theories posit that people tend to explain actions with fixed traits if they believe that personal traits are non-malleable (i.e., entity theory), whereas they are inclined to decipher causes in terms of situational factors if they believe that human attributes are dynamic and malleable (i.e., incremental theory; Chiu et al. [73]; Dweck et al. [74]; Gervy et al. [75]). Kam [76] proposed that implicit beliefs actively shape the effectiveness of trust repair outcomes. According to Kam [76], individuals with entity mindsets are more likely to make internal, stable, and controllable attributions than those with incremental beliefs; trust violations should have more negative impacts on individuals with entity orientation, in contrast to individuals with incremental orientation; and trust repairs are less successful for entity-oriented individuals with slow trust recovery. Therefore, the present study incorporated participants’ entity beliefs as a covariate in HRI trust repairs.

4 Method

4.1 Participants

The current sample was collected from a departmental research pool at a South Korean university ($n = 70$), a crowdsourcing survey platform, Prolific ($n = 65$), and a professional survey company, Macromill Embrain ($n = 233$). The study was approved by the Institutional Review Board (IRB) of the University of Oklahoma (IRB#:12967), and all methods were performed following the relevant guidelines and regu-

lations. This research project was part of a larger study [77] that collected data from US college students. This study was the second component of data collection from South Korean respondents. Data were collected from May 17 to October 20, 2021. As recruitment criteria, the participants were all South Korean citizens aged > 18 years. A total of 368 participants completed the experiments, excluding 18 incomplete attempts. Each participant watched three different scenarios of HRIs, which were randomly assigned out of 13 different conditions (3 failure types $\times$ 4 trust repair strategies + 1 control group) and resulted in a total of 1,027 valid observations that passed attention checks.

Of the total, 179 participants (48.6%) self-identified as males, 182 as females (49.4%), 3 as nonbinary (0.8%), and 4 did not provide gender information (1.0%). Their ages were distributed as: 158 participants were aged between ages 18 and 29 (42.9%); 80 between ages 30 and 39 (21.7%); 60 between 40 and 49 (16.3%); 57 between 50 and 59 (15.5%); and 13 unreported (3.5%). Thus, the median age of the sample was between 30 and 39 years. Participants recruited from the university research pool were granted extra credit as compensation. Whereas those recruited from professional survey companies were compensated with monetary payments; each participant was paid approximately \$3, considering the average completion time and the company's hourly and commission rates.

4.2 Procedure

The entire experimental procedure was conducted online using a Qualtrics interface. First, the original English survey and experimental material were translated into Korean by a native Korean speaker and bilinguals between Korean and English. Subsequently, the translated version was closely examined and revised by two native speakers (also bilinguals). The participants were informed of the general purpose of the study and directed to self-report their propensity to trust robotics and entity beliefs before providing their demographic information. Based on their gender identification, they were assigned to watch a set of pre-recorded videos with either the male or female interactant voice: male participants were matched with the male voice, female participants were matched with the female voice, and nonbinary and unreported participants were randomly assigned to either condition. The purpose of this gender-based voice match was to enhance participants' identification with the human interactant.

The study adopted a within-between subjects design; essentially, the between-subjects difference tests were conducted with participants being assigned to repeated measures. In addition to one failure-free control condition, 12 experimental conditions were considered in total: three failure types (logic, semantic, and failure) $\times$ 4 repair methods (internal attribution apology, external attribution apology, denial, and

on repair). Three observations were acquired from each participant. Therefore, the unit of analysis was each observation rather than an individual. Before the experiment began, participants watched an introductory video about NAO, in which the robot was presented as a medical assistant used in the real world. Next, participants were randomly assigned to three of the aforementioned 13 conditions (3 failure types $\times$ 4 repair methods, and 1 control) with randomized orders. Each experimental condition contained approximately 80 observations, and all conditions added up to 1027 observations.

In each experimental condition, participants watched a short clip (approximately 90 s long) of NAO's interactions with a human, in which NAO answered questions based on a given prescription, made the same type of mistakes twice (but different across conditions), and either attempted to repair trust (with various strategies across conditions) or not. Subsequently, the participants were asked to imagine themselves as interactants in the video they watched and fill a short questionnaire to evaluate their competence-based trust beliefs, integrity-based trust beliefs, perceived severity of violations, and causal attributions after watching each interaction. A manipulation check was conducted by asking participants to identify the number of mistakes made by NAO, and the cases containing incorrect responses ($n = 77$) were excluded from further analysis. We also asked participants to describe in detail what failures NAO had made to remind them of the nature and content of observed failures. In the control condition, participants watched a failure-free video of the same HRI and completed the same measures for competence-based trust beliefs and integrity-based trust beliefs only. In total, each participant watched three video clips, followed by an evaluation questionnaire (see Fig. 1). Table 1 presents the sample script, and the full video scripts can be obtained from the authors upon request.

Although the observations acquired from the same individual were not completely independent, the residuals were randomly distributed because of the two layers of randomization (combination and order). As the residuals were not correlated systematically across observations [78], the design was only a partial violation of the assumption of independent observations.

4.3 Measures

4.3.1 Propensity to Trust Robotics

Dispositional trust precedes learned trust [79]. To assess participants' general propensity to trust robotics, six five-point Likert-type scale items (1 = *Strongly agree*, 5 = *Strongly disagree*; e.g., "I think it's a good idea to rely on robots for help") were adapted from the Propensity to Trust Technology Scale [80]. Scale reliability was evaluated with McDonald's ω , a measure of scale reliability generated by estimations

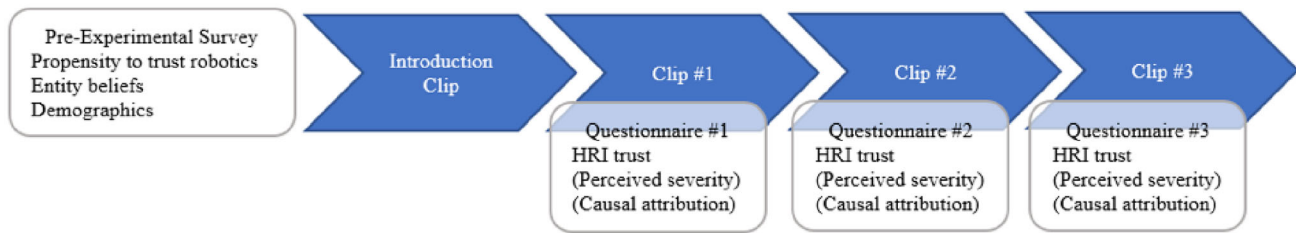


Fig. 1 Illustration of the experimental procedure

Table 1 Sample failures and repair attempts

Sample question: “How often should I use it?”	
<i>Failure types</i>	
Logic	“You should apply it 8 times per day.”
Semantic	“Each gram contains 1 mg of triamcinolone.”
Syntax	“...”
No failure	“You should apply it 2 to 4 times a day.”
<i>Repair Attempts</i>	
Internal-attribution	“I sincerely apologize for this mistake. This is definitely my responsibility. It won’t happen again in the future.”
Apology	“I sincerely apologize for this mistake; but it was not fully my responsibility — there was some disturbance in the environment. It won’t happen again in the future.”
External-attribution	
Apology	
Denial	“I don’t think this is my fault, so I am not the one to blame. My performance shouldn’t be of concern.”
No repair	“...”

of maximum likelihood that is commonly acknowledged to be more reliable than Cronbach’s α [81]. This scale reached acceptable reliability ($\omega = 0.72$), based on the conventional cutoff value of reliability measures (0.70) [82]. The average value of the six measurement items was calculated for the remaining analyses as ($M = 3.20$, $SD = 0.45$).

4.3.2 Entity Beliefs

Six 7-point items (1 = *Strongly disagree*, 7 = *Strongly agree*; [74]) were used to measure entity beliefs in intelligence and morality (i.e., the extent to which participants believe personal intelligence and morality are nonmalleable). The scale was reliable overall, with McDonald’s $\omega = 0.85$. Both the intelligence- ($\omega = 0.90$) and morality- ($\omega = 0.85$) based entity beliefs subscales were highly reliable. Confirmatory factor analysis (CFA) revealed that the bifactor model ($RMSEA = 0.03$, $CFI = 0.99$, $SRMR = 0.02$) provided an excellent fit ($\chi^2 = 10.79$, $df = 8$, $p = 0.21$) compared with the cutoff values proposed by Hu and Bentler [83]: ($RMSEA \leq .06$, $CFI \geq 0.95$, $SRMR \leq 0.08$). The average scores of both the intelligence- ($M = 3.88$, $SD = 1.32$) and morality-based ($M = 3.82$, $SD = 1.36$) entity belief subscales were calculated for further analysis. Additionally, the intelligent- and

morality-based entity beliefs were positively correlated ($\tau = 0.48$, $p < 0.001$).

4.3.3 Post-interaction HRI Trust

Owing to the multifaceted nature of HRI-based trust, trust beliefs (i.e., beliefs that the agent possesses desirable attributes) were investigated on two axes: competence- and integrity-based trust. However, because an HRI trust measure encompassing both axes did not yet exist at the time the study, herein, two scales, each emphasizing one axis, was incorporated to form post-interaction trust measures. The items were selected based on the definition of each dimension as advised by two outside experts. Competence-based post-interaction trust was evaluated using seven 11-point items (1 = 0%, 11 = 100%) with the highest content validity ratios (CVR)¹ values extracted from the HRI-Trust Perception Scale [15], which performed great reliably, $\omega = 0.93$. This scale emphasizes the role of robots’ competence in HRI trust (e.g., “What % of the time will this robot function successfully?”) and the mean score was calculated for further analysis: $M = 5.98$,

¹ $CVR = (n_e - N/2)/(N/2)$, n_e = number of subject matter experts (SMEs) indicating that an item is “extremely important to include in the scale”; N = total number of SMEs.

$SD = 2.16$. Integrity-based post-interaction trust was assessed using seven 7-point Likert items (1 = *Not true at all*, 7 = *Very much true*) adapted from the Checklist for Trust Between People and Automation Scale [84], which also achieved good internal reliability ($\omega = .84$). The selected items stressed integrity-based HRI trust (e.g., “The robot has integrity”), and the mean score was calculated for further analysis: $M = 4.14$, $SD = 1.20$. The two types of trust (competence- and integrity-based) were positively correlated ($\tau = 0.66$, $p < 0.001$).

4.3.4 Severity of Failures

Perceived severity of robotic failures was evaluated by three items taken from Weun et al. [85], such as “If this problem were really happening to me, I would consider the problem to be... (1 = *Not very severe*, 5 = *Very severe*)”. The scale exhibited good reliability according to McDonald’s $\omega = 0.81$, and the average scores were calculated for further analysis: $M = 3.06$, $SD = 1.02$. An additional bivariate correlation analysis revealed that the perceived severity of violations was negatively associated with competence-based ($\tau = -0.63$, $p < 0.001$) and integrity-based ($\tau = -0.61$, $p < 0.001$) post-interaction trust beliefs.

4.3.5 Causal Attributions

Participants’ causal attributions were measured with twelve 10-point bipolar items from the Revised Causal Dimension Scale (CDSII; McAuley et al. [86]), with slightly adjusted phrases to fit the present study context. Altogether, the scale contains four subscales, with one indicating the locus of causality (i.e., whether the cause of robotic failure is located in the situation or the human interactant), one indicating stability (i.e., whether the cause of robotic failures is stable over time), and two indicating controllability. To fully capture the nuances of controllability, CDSII introduced a two-factor structure: external (i.e., whether the cause can be controlled by the NAO robot in the videos) and personal (i.e., whether the cause of robotic failures is under the control of the human interactant) control. Each subscale contains three items, and higher scores on each subscale indicate higher levels of perceived external locus of causality, external uncontrollability, instability, and personal uncontrollability.

The original scale achieved acceptable internal reliability ($\omega = 0.70$). The reliability scores of the subscales are presented in Table 1. The chi-square test of model fit was significant ($\chi^2 = 519.51$, $df = 48$, $p < 0.001$), and the global goodness-of-fit indices of the initial CFA model did not meet the given criteria ($RMSEA \leq 0.06$, $CFI \geq 0.95$, $SRMR \leq 0.08$.) proposed by Hu and Bentler [83]: $RMSEA = 0.10$, $CFI = 0.88$, $SRMR = 0.08$. According to Kline [87], acceptable path coefficients should be greater than 0.6.

Therefore, one item was dropped from the subscale of locus, thus resulting in a two-item solution (Spearman’s $\rho = 0.41$, $p < 0.001$), and stability was reduced to a single-item solution and thus excluded from CFA.

The results revealed that the model fit was considerably improved even though the chi-square test remained significant: $\chi^2 = 156.22$, $df = 17$, $p < 0.001$: $RMSEA = 0.09$, $CFI = 0.96$, $SRMR = 0.05$. As Brown and Cudeck [88] suggested that $RMSEA$ values below 0.10 still indicate acceptable model fit, this model was retained as the final solution. The average scores of subscales and bivariate correlations after model revisions are presented in Table 2.

5 Results

To reiterate, we examine how two independent variables, failure types (i.e., logic, semantic, and syntax) and repair methods (i.e., internal-attribution apology, external-attribution apology, denial, and no repair), impact two dependent variables, post-interaction trust (i.e., competence- and integrity-based trust) and perceived severity of failures, with blame attribution being examined as the mediating mechanism. All dependent variables met the assumption of univariate normality following Osborne’s [89] assumption that skewness and kurtosis with absolute values smaller than one indicate a normal distribution of data. The amount of missing data was minimal. Therefore, mean substitutions were performed. A multivariate analysis of covariance (MANCOVA) revealed that the experimental conditions resulted in significantly lower levels of competence- and integrity-based trust than the failure-free control condition.

5.1 Results of RQ1, RQ2, and RQ3

RQ1 and RQ2 are concerned with the main effects of failure types and trust repair methods on the two types of trust beliefs and the perceived severity of trust violations. After removing three multivariate outliers, we performed MANCOVA with failure types and repair methods as independent variables and competence-based trust, integrity-based trust, and perceived severity of trust violations as dependent variables. The assumption of variance homogeneity was satisfied, but the assumption of covariance homogeneity was violated [90]. To compensate for this violation, Pillai’s Trace scores for the multivariate test results are reported. The propensity to trust robotics ($p < 0.001$), intelligence-based entity beliefs ($p = 0.15$), and morality-based entity beliefs ($p < 0.05$) were considered as covariates.

In response to RQ1, the results revealed that group means significantly differed with failure types at the multivariate level: Pillai’s Trace = 0.05, $F(6, 1908) = 8.71$, $p < 0.001$, partial $\eta^2 = 0.03$. At the univariate level, significant

Table 2 Scale reliability, descriptive statistics and correlations for subdimensions of CDSII

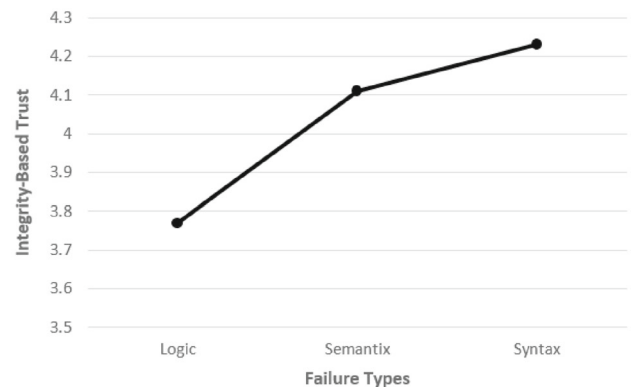
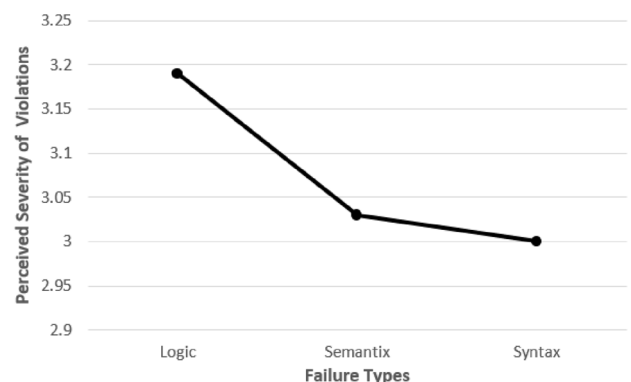
Variables	McDonald's ω	M	SD	1	2	3	4
<i>The initial scale</i>							
1. Locus	0.66	5.97	1.56	–			
2. External control	0.80	4.59	1.77	–0.10**			
3. Stability	0.51	5.78	1.44	0.24**	–0.15*		
4. Personal control	0.85	5.65	2.12	0.67**	–0.06	0.10**	–
<i>The revised scale</i>							
1. Locus	–	5.88	1.73	–			
2. External control	0.80	4.59	1.77	–0.13			
3. Stability	–	6.04	1.96	0.20**	–0.23*		
4. Personal control	0.85	5.65	2.12	.66*	0.06	–0.09**	–

* $p < 0.001$; ** $p < 0.01$

group differences were observed for two dependent variables, integrity-based trust ($F(2, 18.49) = 36.98, p < 0.001$, partial $\eta^2 = 0.03$), and perceived severity of trust violations ($F(2, 3.12) = 6.25, p < 0.05$, partial $\eta^2 = 0.01$). Two significant mean differences were observed in integrity-based trust (-0.35 , 95% CI $[-0.51, -0.18]$) between logic ($M = 3.77, SE = 0.06$) and semantic ($M = 4.11, SE = 0.06$) failures, thus indicating that logic failures resulted in lower levels of integrity-based trust compared with semantic failures ($p < 0.001$). Similarly, the mean difference (-0.47 , 95% CI $[-0.63, -0.30]$) was significant between logic and syntax ($M = 4.23, SE = 0.06$) failures, thus indicating that logic failures also resulted in lower levels of integrity-based trust compared with syntax failures ($p < 0.001$).

For perceived severity of trust violations, a significant group mean difference (0.16 , 95% CI $[0.01, 0.30]$) was observed between logic ($M = 3.19, SE = 0.05$) and semantic ($M = 3.03, SE = 0.05$) failures, thus implying that logic failures were perceived as more severe than semantic failures ($p < 0.05$). Additionally, a significant group mean difference was observed in perceived severity of trust violations (0.19 , 95% CI $[0.04, 0.33]$) between logic and syntax ($M = 3.00, SE = 0.05$) failures, thus implying that logic failures were perceived as more severe than syntax failures ($p < 0.05$). The findings for the two dependent variables exhibited a consistent pattern. In general, logic failures led to the worst outcomes among the three types of failures (see Figs. 2 and 3 to visualize the estimated means).

In response to RQ2, the analysis revealed that the group means significantly differed by the repair type at the multivariate level: Pillai's Trace = 0.21, $F(9, 2865) = 24.16, p < 0.001$, partial $\eta^2 = 0.07$. At the univariate level, group means differed for all three dependent variables: competence-based trust, $F(3, 955) = 20.87, p < 0.001$, partial $\eta^2 = 0.06$; integrity-based trust, $F(3, 955) = 47.08, p < 0.001$, partial $\eta^2 = 0.13$; and perceived severity of trust violations, $F(3,$

**Fig. 2** Estimated Means of Integrity-Based Trust (calculated by averaging seven 7-point Likert items adapted from Jian et al. [84]) by Robotic Failure Types. Standard error for three conditions (from left to right) is estimated as follows: 0.06, 0.06, 0.06**Fig. 3** Estimated Means of Perceived Severity of Violations (calculated by averaging three 5-point Likert items adapted from Weun et al. [85]) by Robotic Failure Types. Standard error for three conditions (from left to right) is estimated as follows: 0.06, 0.06, 0.06

955) = 37.52, $p < 0.05$, partial $\eta^2 = 0.11$. The estimated mean values per group are presented in Table 3.

For competence-based trust, the mean difference was significant between (a) internal and external attribution apology

Table 3 Estimated means of dependent variables by repair methods

Dependent Variables	Repair types	<i>M</i>	<i>SE</i>	95% Confident interval	
				Lower Bound	Upper Bound
Competence based trust	Apology (Internal)	6.51	.12	6.27	6.75
	Apology (External)	5.92	.12	5.68	6.16
	Denial	5.15	.13	4.90	5.41
	No repair	5.56	0.13	5.30	5.81
Integrity based trust	Apology (Internal)	4.55	0.07	4.42	4.69
	Apology(External)	3.95	0.07	3.82	4.08
	Denial	3.43	0.07	3.29	3.57
	No repair	4.21	0.07	4.07	4.36
Perceived Severity of	Apology (Internal)	2.58	.06	2.47	2.70
	Apology (External)	2.97	0.06	2.86	2.86
Trust	Based trust	3.39	0.06	3.27	3.27
Violation	No repair	3.34	0.06	3.21	3.21

Covariates appearing in the model are evaluated at the following values: propensity to trust robots = 3.43, intelligence-based entity beliefs = 3.88, and morality-based entity beliefs = 3.82

($p < 0.001$); (b) internal attribution apology and denial ($p < 0.001$); (c) internal attribution apology and no repair ($p < 0.001$); (d) external attribution apology and denial ($p < 0.001$); (e) external attribution apology and no repair ($p < 0.05$); and (f) denial and no repair ($p < 0.05$). Overall, the internal attribution apology elicited higher levels of competence-based trust than the external attribution apology, followed by no repair or denial.

For integrity-based trust, a significant group mean difference was observed between (a) internal and external attribution apology ($p < 0.001$); (b) internal attribution apology and denial ($p < 0.001$); (c) internal attribution apology and no repair ($p < 0.001$); (d) external attribution apology and denial ($p < 0.001$); (e) external attribution apology and no repair ($p < 0.01$); and (f) between denial and no repair ($p < 0.001$). The results indicated that the internal attribution apology led to higher levels of integrity-based trust, followed by no repair, external attribution apology, and denial.

For perceived severity of violations, group means significantly differed between internal attribution apology and external attribution apology ($p < 0.001$); (a) internal attribution apology and denial ($p < 0.001$); (b) internal attribution apology and no repair ($p < 0.001$); (c) external attribution apology and denial ($p < 0.001$); and (d) external attribution apology and no repair ($p < 0.001$). However, no difference was observed between denial and no repair. Thus, denial and no repair were considered the most serious trust violations, followed by external attribution apology, and internal attribution apology achieved the best outcome in mitigating the negativity caused by trust violations.

The third research question explored the potential interaction effects between failure types and trust repair strategies. At the omnibus level, the interaction was statistically signif-

icant: Pillai's Trace = 0.05, $F(18, 2865) = 2.67$, $p < 0.001$, partial $\eta^2 = 0.02$. However, no significant group difference was found in competence-based trust ($p = 0.23$), integrity-based trust ($p = 0.33$), or perceived severity of trust violations ($p = 0.07$) at the univariate level. No specific combinations between failure types and repair methods resulted in any special effects on trust repair and perceived severity of trust violations.

5.2 Results of H1, H2, and H3 Tests

Three linear regression analyses were performed to test H1, H2, and H3, which proposed that internal, unstable, and externally uncontrollable and/or personally controllable causal attributions in HRI are associated with greater competence-based trust beliefs (H1), greater integrity-based trust beliefs (H2), and less perceived severity of trust violations (H3). The first regression model predicting competence-based trust belief by perceived external locus of causality, non-external control, instability, and non-personal control was significant (adjusted $R^2 = 0.09$, $p < 0.001$). Perceived external locus of causality ($\beta = -0.12$, $p < 0.01$), external uncontrollability ($\beta = -0.08$, $p < 0.01$), and personal uncontrollability ($\beta = -0.14$, $p < 0.001$) were negatively associated with competence-based trust, whereas instability ($\beta = 0.20$, $p < 0.001$) was positively associated with competence-based trust. This implies that the more the participants felt the causes of failures to be internal to themselves, unstable, and controllable by the human interactant, the more trust they assigned to the robot's competence after the interaction, which fully supports H1.

The model predicting integrity-based trust beliefs was also significant (adjusted $R^2 = .11$, $p < 0.001$). The results revealed two significant predictors: instability ($\beta = 0.19$,

$p < 0.001$) and personal uncontrollability ($\beta = -0.31$, $p < 0.001$), which indicated that participants ascribed more integrity-based trust to the robot when they perceived the causes of failures to be more unstable and personally controllable. Overall, the results support H2b and H2c but not H2a.

The third model predicting perceived severity of trust violations with blame attributions revealed that perceived external locus ($\beta = 0.14$, $p < 0.001$) and personal uncontrollability ($\beta = 0.02$, $p < 0.05$) were positively associated with perceived severity of trust violations, whereas lower perceived instability ($\beta = -0.21$, $p < 0.001$) led to more perceived severity. This implies that participants assessed the robot's failures to be more severe if they thought the causes were external to them and uncontrollable by them, as well as stable over time when they imagined themselves as the human interactant. This result supports H3.

5.3 Results of RQ4, RQ5, and RQ6

Further, eight multivariate outliers ($p < 0.001$) were removed and another MANCOVA was conducted to explore the impact of failure types and repair methods on blame attributions under univariate homogeneity of variance, with propensity to trust robotics ($p < 0.001$), intelligence-based entity beliefs ($p = 0.10$), and morality-based entity beliefs ($p < 0.05$) as covariates. Because the assumption of homogeneity of covariance was violated, Pillai's Trace scores were reported.

In response to RQ4, the analysis revealed that group means significantly differed with the failure type at the multivariate level: Pillai's Trace = 0.03, $F(8, 1894) = 3.33$, $p < 0.01$, partial $\eta^2 = 0.01$. At the univariate level, the group means significantly differed in external uncontrollability, $F(2, 958) = 6.25$, $p < 0.05$, partial $\eta^2 = 0.02$. Table 4. presents the estimated means. Pairwise comparisons revealed that the mean difference in external uncontrollability was significant between logic and semantic failures ($p < 0.001$). Similarly, the mean difference in external uncontrollability was significant between logic and syntax failures ($p < 0.05$). The results indicated that logic failures were perceived as more controllable by the robot compared with semantic and syntax failures.

For RQ5, the analysis revealed that the main effects were significant for repair types (Pillai's Trace = 0.06, $F(12, 2844) = 5.08$, $p < 0.001$, partial $\eta^2 = 0.02$), and the group means significantly differed with personal uncontrollability ($F(3, 958) = 12.40$, $p < 0.001$, partial $\eta^2 = 0.04$). The results for the estimated means are presented in Table 5. Significant differences were observed in four comparisons: the mean difference between (a) internal-attribution apology and denial ($p < 0.001$); (b) external attribution apology and denial ($p < 0.05$); (c) external attribution apology and no repair

($p < 0.001$); and (d) denial and no repair ($p < 0.001$). The results indicate that denial led to the highest levels of perceived personal uncontrollability, followed by external attribution apology, internal attribution apology, and no repair.

The interaction between failure type and repair strategy was significant at the omnibus level (Pillai's Trace = 0.04, $F(24, 3796) = 1.57$, $p < 0.05$, partial $\eta^2 = 0.01$) but not at the univariate level for external locus ($p = 0.97$), external uncontrollability ($p = 0.30$), instability ($p = 0.06$), and personal uncontrollability ($p = 0.16$). Hence, failure types and repair strategies did not seem to interact to influence blame attribution. Please see the Table 6 provides a summary of all the key findings and results.

6 Discussion

Focusing on the relationship between user intentions and machine-generated output, this study tested a new typology of performance failures (namely, logic, semantic, and syntax failures) in HRI. The typology is human-centered in that it focuses on the purpose, process, and outcome of HRI, particularly aiming at making the different failure types identifiable and distinguishable by lay users. Moreover, it can be applied to other human-machine communication (HMC) contexts than HRI, such as human-computer, human-automation, and human-agent interactions, because it distinguishes failures by different mismatches between user goals and erroneous practices (i.e., expected outcomes and actual outcomes). Its cross-contextual applicability can bridge some gaps in context-specific failure typologies and help reduce the discrepancy between theorization and investigation of HMC failures. Although numerous failure typologies exist, only few of them effectively guide research designs and interpretations of empirical findings.

Furthermore, this study explored the effects of failure types and repair methods (i.e., internal and external attribution apology, denial, and no repair) on users' trust beliefs and perceptions of trust violation severity, extending previous research on human-human and human-machine trust repair. It unveils some nuances that distinguish human-to-machine trust from human-to-human trust beyond the parallels between the two types of trust. The implications of the research findings are discussed in more detail as follows.

First, the findings of this study facilitate an understanding of the nature of common trust violations in HRI. Unlike Sebo et al.'s design [43], in which the robot deliberately framed failures as competence- or integrity-based violations, the present study did not explicitly frame the failures. Instead, this study explored how people perceive basic performance failures without external framing or interpretation. The results revealed that the different types of failures varied

Table 4 Estimated means of attribution dimensions by failure types

Dimensions	Failure types	<i>M</i>	<i>SE</i>	95% Confident interval	
				Lower bound	Upper bound
Locus	Logic	5.87	0.10	5.68	6.06
	Semantic	5.85	0.10	5.66	6.04
	syntax	5.92	0.10	5.74	6.11
External control	Logic	4.31	0.10	4.12	4.50
	Semantic	4.85	0.10	4.66	5.04
	syntax	4.59	0.10	4.40	4.78
Stability	Logic	6.13	0.11	5.93	6.34
	Semantic	6.02	0.11	5.81	6.23
	syntax	5.98	0.11	5.77	6.18
Personal control	Logic	5.84	0.11	5.62	6.07
	Semantic	5.46	0.11	5.24	5.69
	syntax	5.68	0.11	5.46	5.90

Higher scores on each dimension indicates higher levels of external locus, external uncontrollability, instability, and personal uncontrollability. Covariates appearing in the model are evaluated at the following values: propensity to trust robots = 3.43, intelligence-based entity beliefs = 3.88, and morality-based entity beliefs = 3.82

Table 5 Estimated means of attribution dimensions by repair methods

Dimensions	Repair types	<i>M</i>	<i>SE</i>	95% Confident Interval	
				Lower bound	Upper bound
External locus	Apology (Internal)	5.74	0.11	5.52	5.96
	Apology (External)	5.87	0.11	5.65	6.09
	Denial	6.08	0.12	5.85	6.31
	No repair	5.83	0.12	5.60	6.06
External Uncontrollability	Apology (Internal)	4.55	0.11	4.34	4.80
	Apology (External)	4.49	0.11	4.28	4.70
	Denial	4.55	0.12	4.33	4.78
	No repair	4.75	0.12	4.52	4.97
Instability	Apology (Internal)	6.30	0.12	6.07	6.54
	Apology (External)	5.96	0.12	5.72	6.20
	Denial	6.04	0.13	5.80	6.29
	No repair	5.87	0.13	5.62	6.12
Personal Uncontrollability	Apology (Internal)	5.47	0.13	5.22	5.72
	Apology (External)	5.80	0.13	5.54	6.05
	Denial	6.26	0.14	6.00	6.52
	No repair	5.13	0.14	4.86	5.40

Higher scores on each dimension indicates higher levels of external locus, external uncontrollability, instability, and personal uncontrollability. Covariates appearing in the model are evaluated at the following values: propensity to trust robots = 3.43, intelligence-based entity beliefs = 3.88, and morality-based entity beliefs = 3.82

primarily according to participants' perceptions of robotic integrity, thus indicating that participants fundamentally distinguished between different performance failures in terms of integrity issues. The existence of any failure raised concerns regarding the robot's competence compared with the control condition (i.e., no failure). This finding is particularly interesting as it indicates that with highly intelligent machines, participants are less doubtful of their competence but more concerned about their integrity. Prior research

achieved mixed findings concerning the effects of perceived robotic deceptions on human appraisal [91], thus reflecting the possibility of a trade-off between intelligence- and integrity-based trust. However, the current study suggests that integrity is the more salient determinant of HRI trust, and the gains in perceived competence building vis-a-vis the loss of integrity, if any, cannot compensate for its negativity.

Evidently, logic failures were perceived as more detrimental than semantic or syntax failures for integrity-based

Table 6 Summary of findings and results

RQs and Hs	Findings and results
<i>RQ1</i> How do different types of performance failures impact dependent variables?	Logic failures had more negative impacts on integrity-based trust beliefs and perceived severity of violations than semantic and syntax failures
<i>RQ2</i> How do different types of trust repair attempts impact dependent variables?	Both internal- and external-attribution apology outperformed no repair and denial, and internal-attribution apology outperformed external-attribution apology
<i>RQ3</i> How do failure types and trust repair methods interact to influence dependent variables?	There was no significant interaction
<i>H1</i> After failure occurrence, competence-based trust toward robots will be better repaired when more (a) internal, (b) unstable, and (c) externally uncontrollable and/or personally controllable causal attributions.	H1 was fully supported. Perceived internal locus, instability, and controllability by both the human and robot were positively associated with competence-based trust
<i>H2</i> After failure occurrence, integrity-based trust toward robots will be better repaired when more (a) internal, (b) unstable, and (c) externally uncontrollable and/or personally controllable causal attributions.	Only H2b and H2c were supported. Controllability by the human and instability were positively associated with integrity-based trust
<i>H3</i> After failure occurrence, perceived severity of violations by robots will be lower when more (a) internal, (b) unstable, and (c) externally uncontrollable and/or personally controllable causal attributions.	H3 was fully supported. Perceived internal locus, instability, and controllability by the human were positive associated with perceived severity of violations
<i>RQ4</i> How do different types of performance failures impact blame attributions?	Logic failures were perceived with more robot controllability than semantic and syntax failures
<i>RQ5</i> How do different types of trust repair attempts impact blame attributions?	Denial was perceived with the least personal controllability, followed by external- and internal-attribution apology and no repair
<i>RQ6</i> How do failure types and trust repair methods interact to influence blame attributions?	There was no significant interaction

trust and perceived severity of trust violations. The results indicated that partially correct responses of the robot were not perceived as an indicator of more competence than completely wrong responses or no responses but as an indicator of less integrity. This finding is counterintuitive in that the relationship between levels of output accuracy and competence-based trust is not linear. Rather, the judgment is in line with the logic of “all or nothing.” Possibly, people prefer failures to be as easily detected as possible, and a seemingly reasonable but incorrect answer entails greater risk of intentional deception. This potential scenario resonates with the overpowering concerns for robotic integrity as their intelligence levels greatly advance.

Unexpectedly, the interaction between violation types and repair strategies differed in the present study from that of previous research [11, 43]. For every failure type, the internal attribution apology outperformed denial in boosting competence- and integrity-based trust beliefs and mitigated perceived severity of trust violations, regardless of whether the failure was considered more or less of an integrity-based trust violation than others. This finding is in stark contrast to previous observations in human-human and human-machine trust repair research. Based on prior studies, denial should achieve better trust repair outcomes compared with internal

attribution apology when a robot violates people’s expectations of its integrity. However, this finding is inconsistent with the predicted interactions between violation types and repair methods by Kim et al. [11] and potentially demonstrates that the ontological differences between humans and machines can lead to different trust repair patterns. Perhaps denial from robots is less acceptable for people than that from humans, because social robots are supposed to be humble servants of humans; thus, an apology is more aligned with the values in embedded robotics, regardless of violation types. Possibly, humans are more concerned about robotic deceptions rather than human deceptions because of the prolonged fear of robots intentionally tricking humans, which is congruent with the finding in the present study that integrity was emphasized above and beyond competence in the context of robotic trust repairs.

The study additionally explored another variant of apology, the external attribution apology, which exerted complex influences on user trust. Internal attribution apology appeared to be more effective than the external-attribution apology, whereas denial was, in general, the worst trust repair strategy. Further, internal attribution apology was more effective in reducing both integrity-related concerns and perceived severity of trust violations compared with no repair, whereas exter-

nal attribution apology achieved mixed outcomes. External attribution apology elicited lower levels of integrity-based trust compared with no repair, but it also led to lower levels of perceived severity of violations compared with no repair. Therefore, the importance of the trade-off between integrity-based trust repair and perceived severity of trust violations must be determined before employing an external attribution apology.

Analyses of blame attribution further elucidated the underlying mechanisms of trust violations and repairs in HRI, and the results of our hypotheses testing were consistent with those of previous blame attribution studies [5, 12, 41]. Notably, competence-based trust was positively associated with both external and personal controllability, in addition to internal locus and instability. This is in contrast to the extant belief in human-to-human trust literature that if the cause of blame is deemed controllable by the trustee, the trustor will judge the mistake more harshly. When participants felt that performance failures were jointly controlled by both the human interactant and robot, they perceived the robot to be more trustworthy in terms of its competence. This indicates that people value both the autonomy and controllability of robots, which can provide practical insights for the development of future robotic systems. For integrity-based trust, locus of controllability was surprisingly not a significant predictor, one explanation for which could be that locus did not affect participants' evaluations of robotic integrity because it only involved participants' internal qualities and situational factors, relatively independent from the robot. Conversely, perceived severity of trust violations was related to locus, stability, and controllability. External uncontrollability was significantly associated with only one of the variables of interest, whereas personal uncontrollability was linked to all. This may indicate that people care more about whether they have the power to control the situation when a robot makes mistakes rather than whether the robot holds controllability. This finding highlights that humans need to be in a position of control in HRI and thus is in agreement with the ongoing HMC research in explainable agency [92], through which humans obtain a better grasp of the operation mechanisms of intelligent machines, in addition to the benefits of increasing mechanical controllability and navigability.

The findings of group differences in blame attributions corresponded with the results of the two types of trust beliefs and perceived severity of violations. Logic failures were considered more controllable by the robot than the other two failure types, which could explain why logic failures were also deemed the most dishonest and severe of all. On the other hand, personal uncontrollability helps explain why denial was the most harmful approach. Essentially, when the robot denied responsibility, expressed no repentance, and made no promise for future improvement, participants felt things were out of their control. However, interestingly, not implement-

ing any repair strategy elicited the highest levels of perceived personal controllability. When the robot takes no action after trust violations, people seem to develop the strongest feeling that everything is under their control. This may again be an indicator of the innate fear of artificial intelligence and robotics. Although trust repairs entailed some positive influences on HRI, participants perceived less control in the situation, most likely owing to the additional agency displayed by the robot in the repair scenarios. Therefore, humans' sense of control must be urgently restored through robotic designs as intelligent machines' autonomy increases.

6.1 Implications

The new failure typology tested in this study can enhance the understanding and exploration of human perceptions of machinery failures in future research. The findings concerning failure types and repair methods can potentially benefit human-centered designs in HRI and other relevant contexts, such as human-computer interactions, human-automation interactions, and human-agent interactions, by identifying the optimal trust repair strategy. Overall, developers and designers of upcoming technology need to prioritize reducing logic failures in HRI and make robots completely assume responsibility when performance failures occur. In particular, the findings emphasize the importance of establishing a robot's integrity. In addition to the advancement in machine intelligence and learning capacity, technology must be perceived as upholding integrity and benevolence, be it from its own or its designers.

6.2 Limitations and Future Directions

The online experimental design may have introduced some generalizability issues in the current study. As the participants did not interact with the embodied robot directly in a physical environment, they might have evaluated the robot differently because of lower levels of interactivity and social presence [93]. As they observed only the interactions, participants might have attributed fewer faults internally, although they were asked to imagine as if they had been in those situations themselves. Because internalization of the experimental scenarios is influenced by participants' identification with the focal person in videos they watched, generalizability of our conclusions is limited and participants' attribution patterns may alter without direct interactions with the robot even with repeated psychological inducement. Additionally, the social nature of robotics may not have been as salient because of the lack of actual interactions. In response to this limitation, future research should consider replicating this study under improved conditions. Conversely, the between-within subjects designs might have introduced some undesirable statistical biases despite our verification of nonexistent auto-

correlations via Durbin-Watson tests and the independence of observed errors via residual plots. The between-group differences could have been larger if a between-subjects design had been adopted. Additionally, the participants might have had difficulties judging the stability of the given robotic failures because the in-lab designs entailed no interaction history or anticipated future contact. Hence, future research can introduce time lapses and explore the effects of robotic trust repairs in the long term.

Future research could investigate other trust repair methods for robots, including but not limited to oral accounts, such as providing external support and offering compensation to human interactants. Another factor worth examining is linguistic agency assignment (i.e., the ascription of an action to an entity; Bell et al. [94]). As robotics has higher embodiment levels and more autonomy than other AI-based machines such as intelligent virtual assistants (e.g., Alexa and Siri), comparing the performance failures and repair types of such robotic technologies with those of other technologies without physical embodiment or with weaker AI can present interesting results. Finally, cross-cultural comparisons can provide additional insights, given the importance of cultural factors in trust in robots [79, 95, 96].

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Declarations

Conflict of interests The authors declare that they have no other conflict of interest. The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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